

REPORT CLASSIFICATION

Classification Type	Full Form Analysis
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DATE / TIME

Date of Occurrence	2023-01
Local Time Of Day	0001-0600

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Flight Conditions	IMC
Visibility	2.5
Light	Night
Ceiling	500

AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737-800
Crew Size	2
Operating Under FAR Part	Part 121
Flight Plan	IFR
Mission	Passenger
Flight Phase	Landing
Route In Use	Vectors

COMPONENT 1

Aircraft Reference	X
Aircraft Component	Cockpit Window

Problem	Design
Problem	Malfunctioning

PERSON 1

Aircraft Reference	X
Location In Aircraft	Flight Deck
Reporter Organization	Air Carrier
Function - Flight Crew	First Officer
Function - Flight Crew	Pilot Not Flying
Qualification - Flight Crew	Air Transport Pilot (ATP)
Qualification - Flight Crew	Multiengine
Qualification - Flight Crew	Instrument
Experience - Flight Crew Last 90 Days	160 hrs
Experience - Flight Crew Type	160 hrs
ASRS Report Number	1962553
Human Factors	Communication Breakdown
Human Factors	Confusion
Human Factors	Situational Awareness
Communication Breakdown - Party1	Flight Crew
Communication Breakdown - Party2	Flight Crew

EVENTS

Anomaly	Aircraft Equipment Problem - Critical
Anomaly	Deviation - Altitude - Excursion From Assigned Altitude
Anomaly	Deviation - Altitude - Undershoot
Anomaly	Deviation / Discrepancy - Procedural - Published Material / Policy
Anomaly	Inflight Event / Encounter - Weather / Turbulence
Anomaly	No Specific Anomaly Occurred - Unwanted Situation
Detector - Person	Flight Crew
Were Passengers Involved In Event	No
When Detected	In-flight
Result - General	Flight Cancelled / Delayed

Result - General	Maintenance Action
Result - Flight Crew	Overcame Equipment Problem
Result - Aircraft	Aircraft Damaged

ASSESSMENTS

Contributing Factors / Situations	Aircraft
Contributing Factors / Situations	Human Factors
Primary Problem	Aircraft

NARRATIVE 1

The Company has put out several communications lately about the hydrophobic coating on the windshields and I remain unsatisfied with their explanation and continue to believe that their unwillingness to treat the windshields poses a serious and unnecessary risk to the Safety of our operation. My last flight highlighted the issues. Landing on XL in ZZZ, the ATIS was reporting +TSRA and 2 1/2 miles visibility. We were cleared for the ILS Runway XR. At minimums, the Pilot Flying (PF) announced "landing". While the runway was somewhat distorted by the rain piling up on the windshield, neither of us felt it was bad enough to call for the go-around. Here lies the problem with not treating the windshields with a hydrophobic coating. The crew is lulled into a false sense of security to continue the landing. However, I believe, based on experience, that the Pilot's depth perception is severely and unknowingly impaired by the distorted view.

This was just what we experienced. Just before touchdown, we got a Sink Rate Aural Warning. It was too late in the flare to initiate a go-around, as we touched down almost immediately after the warning. As you might imagine, it was a hard landing. What was unnerving to us both was that neither of us had the slightest inclination that the aircraft was sinking. I attribute this to our distorted view through the windshield. We felt that we had adequate visual reference to continue the landing, but clearly, we unknowingly did not. I have seen this happen other times as well and always in heavy rain. I happen to know that other airlines maintain their windshields better than we do. When I observed (other carrier) land immediately after us in the same conditions, they had a much more normal touchdown. I have worked for operators, in the past, who maintained the hydrophobic coating and never had any issues landing in heavy rain. We also didn't even have windshield wipers on our aircraft. I strongly believe Company is assuming unnecessary risk by choosing to operate this way.

SYNOPSIS

B737-800 First Officer reported vision problems caused by the hydrophobic coating on the windscreens caused misjudgments and an excessive sink rate during landing. A hard landing was entered into the log book for maintenance action.

